

Curro Hillcrest Service tracker

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Grade 12

2022

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Summary

Write a short paragraph describing the scenario for which your program is going to be written. Briefly describe what entities your program will manage:
I have been requested by my school, Curro Hillcrest High school, to design an application that will allow them to manage the number of hours learners have spent participating in the service portfolios, electronically. The application will allow admins (Teachers) to create events and students to log their participation in the event. Each event will be assigned a set number of hours which will be allocated to the learner when they log their participation in the event, enabling learners to track their total hours for each portfolio, which will accumulate and eventually earn them awards as per the school requirements. Learners and teachers will be required to log into the system as the information is sensitive and admins have different access to information and permissions to that of learners. In this application, information relating to the events, users and tracking of events will be stored in a database. The application will allow students to log participation, edit their information and delete the record of their participation in an event. It will allow admins (Teachers) to add, edit, and view all information contained in the database.

0 – No summary or completely inadequate	1 – Summary partially done	2 – Summary encompasses all aspects of the problem

Motivation and research

A brief discussion of similar projects as a result of research. Research is correctly referenced and cited. A motivation is included for the project indicating how this project differs from existing projects. This section should NOT be longer than a page.
Connecteam is a free shift tracking system that allows users to make shift schedules and edit them online in an easy manner. (Connecteam.com, 2019) My project will differ from this as it will not have equal length sifts on a regular basis, but rather events of varying length on inconsistent days. Other, similar systems exist to track employee work time but do not work on an event-based system. I decided to design a project for Curro Hillcrest High school because, as a learner who has participated in many service portfolios, I have identified that the school's system for tracking service hours is old fashioned and often results in missing records and excessive administration work.

0 – No motivation or research provided	1/2 – Similar projects are described, cited and referenced. A motivation is supplied to include	3 - Similar projects are described, cited and referenced. A motivation is supplied to include an

	an explanation of how the project will differ from existing projects.	explanation of how the project will differ from previous projects.

Specifications of Program Function

List each function your program will be able to perform.

The application designed for Curro Hillcrest High will be able to do the following:

Screen 1 (Log-in):

- Allow users to log in using a username and password to access an admin or learner account
- Allow users to press a button that will direct them to the sign-up page where they can sign up for a learner account
- Allow users to select the help button

Screen 2 (Sign-up):

- Allow learners to register/sign up by entering a username and by entering the same password twice
- Allow learners to enter their full name
- Allow learners to enter their grade
- Allow learners to enter their class
- Allow learners to select a gender
- Allow learners to select portfolio options
- Allow learners to select a button to create their account provided all the above information has been entered
- Allow users to select the help button

Screen 3 (Home):

- Allow learners to view their details (name, surname, username, grade, class, gender and portfolios with their accumulated hours in each portfolio)
- Allow learners to select an “Edit account” button
- Allow learners to navigate to other parts of the program through tabs
- Allow users to select the help button

Edit Account (Student):

- Allows users to edit their existing details, in separate fields (Name, surname, grade, class, portfolios and gender)
- Allow users to select an “Enter” button to confirm the changes
- Allow users to select a “Cancel” button to exit the edit screen and return to the home screen

Screen 4 (Events):

- Allow learners to view existing events related to their portfolio selection
- Allow learners to search for events by date which will return event ID, event name, event date, portfolio and how long the event took (Hours)
- Allow learners to search for events by name which will return event ID, event name, event date, portfolio and how long the event took (Hours)
- Allow learners to search for events by ID which will return event ID, event name, event date, portfolio and how long the event took (Hours)

- Allow learners to navigate to other parts of the program through tabs
- Allow users to select the help button

Screen 5 (Participation):

- Allow learners to log their participation in events
- Allow learners to view records of their participation in events
- Allow learners to delete record of their participation in events
- Allow learners to search for events that they have participated in, by portfolio which will return the event ID, the event name, Student ID, hours, and portfolio.
- Allow learners to search for events that they have participated in, by event name which will return the event ID, the event name, Student ID, hours, and portfolio.
- Allow learners to search for events that they have participated in, by event ID which will return the event ID, the event name, Student ID, hours, and portfolio.
- Allow learners to navigate to other parts of the program through tabs
- Allow users to select the help button

Screen 6(Home admin):

- Allow admins to view their details (Name, surname and portfolios)
- Allow admins to select an “Edit account” button
- Allow admins to select a “Create new admin account” button
- Allow admins to navigate to other parts of the program through tabs
- Allow users to select the help button

Edit Admin screen

- Will allow admins to edit their existing details in separate fields (Name, surname, and portfolios)
- Allow admins to select an “Enter” button to confirm any changes they have made
- Allow admins to select a “Cancel” button to exit the screen and discard any changes

Create new admin account screen

- Allow an existing admin to enter details for a new admin (Full name, username, password, confirmation of password, Portfolios)
- Allow admin to select an “Enter” button to create the new admin account
- Allow the admin to select a “cancel” button to exit the screen and discard any information they have entered.

Screen7(Participation admin):

- Allow admins to view all participation records related to their portfolios
- Allow admins to search for participation records by Event ID which will return the event ID, the event name, the student ID, student name, the relevant portfolio and the number of hours for which the event counts.
- Allow admins to search for participation records by event name which will return the event ID, the event name, the student ID, student name, the relevant portfolio and the number of hours for which the event counts.
- Allow admins to search for participation records by student name which will return the event ID, the event name, the student ID, student name, the relevant portfolio and the number of hours for which the event counts.
- Allow admins to search for participation records by portfolio which will return the event ID, the event name, the student ID, student name, the relevant portfolio and the number of hours for which the event counts.
- Allow admins to delete participation records
- Allow admins to create participation records

- Allow admins to navigate to other parts of the program through tabs
- Allow users to select the help button

Screen 8(Events admin):

- Allow admins to view existing events related to their portfolio
- Allow admins to select a “create new event” button
- Allow admins to select an “edit existing event” button
- Allow admins to search for events, by date which will return event ID, event name, event date, portfolio and how long the event took
- Allow admins to search for events, by name which will return event ID, event name, event date, portfolio and how long the event took
- Allow admins to search for events, by event ID which will return event ID, event name, event date, portfolio and how long the event took
- Allow admins to navigate to other parts of the program through tabs
- Allow users to select the help button

Edit Event Screen:

- Will initially present the admin with a single space to enter an event ID
- Will initially present the admin with a “Find event” button to search for an event matching the ID they have entered.
- Will allow the admin to select an “Enter” button to confirm the changes they have made
- Will allow the admin to select a “cancel” button to exit the screen and discard any changes they have made.
- After an event matching an entered ID has been found, the rest of the screen will be displayed with fields containing the current values assigned to that event
- Will allow the admin to change the existing values assigned to an event.

New Event Screen:

- Will allow the admin to enter all the details of an event (name, hours, date, and portfolio)
- Will allow the admin to select an “Enter” button to create the event provided all the fields are complete
- Will allow the admin to select a “cancel” button to exit the screen and discard any details they have entered

Screen 9(Students):

- Allow admins to search for learners by name which will return student ID, student name, grade, class, gender (M/F), catering hours, photography hours, AV hours and first aid hours
- Allow admins to search for learners by grade which will return student ID, student name, grade, class, gender (M/F), catering hours, photography hours, AV hours and first aid hours
- Allow admins to search for learners by Student ID which will return student ID, student name, grade, class, gender (M/F), catering hours, photography hours, AV hours and first aid hours
- Allow admins to navigate to other parts of the program through tabs
- Allow users to select the help button

Screen 10(Help):

- Allow users to select a button that relates to their problem which will direct them to a video that explains how to solve the problem or perform the task
- Allow users to contact someone if none of the solutions provided could assist them

The application will not:

- Allow learners to delete their profile as this would cause errors in the system and the

profiles should stay active after students have left.
- Allow learners to change their username because this information should be constant and not change, as set by the school (usually)
- Allow learners to change their password because this information should be constant and not change
- Allow learners to create events as they do not have permission to do this and only admins should have this ability to prevent errors and overlaps
- Allow learners to edit events as they do not have permission to do this and only admins should have this ability to prevent errors and overlaps
- Allow learners to delete events as they do not have permission to do this and only admins should have this ability to prevent errors and overlaps
- Allow admins to delete their profile as this could cause errors and admin accounts should stay active even after a staff member has left
- Allow admins to change their username because this information should be constant and not change
- Allow admins to change their password because this information should be constant and not change
- Allow new admin accounts to be created outside of an existing admin account as there is no way to ensure that a student or other person is not creating a false account to falsify records
- Allow admins to delete student records as this would cause errors in the system and the profiles should stay active after students have left.
- Allow admins to edit student records because this would cause errors in the system and the profiles should stay active and no information should change.
- Allow students to view other student's details as this information should be kept private.
- Allow users to access information related to portfolios that they do not participate in as it would create unnecessary clutter of records that are not relevant to the user.

0 – No Functions listed	1 – Function list is a single line statement/a list of 4 or less points	2 – Function list is a substantial list of appropriate outcomes	3 – Function List is complete and detailed

Specifications of User Interface

Describe the general look and feel of your program (short paragraph). **List** each screen that you will need and give a brief description (one sentence) of their purpose.

The application will use the colour scheme of the Curro Hillcrest, navy blue and white, and will use a formal, easy to navigate layout. and will provide the following interfaces:

Screen 1 (Login): This screen will have a simple yet formal layout, using the logo of the school in the top left-hand corner. It will allow students to log into a normal user account or it will allow teachers to log into the admin account by entering a unique username and password into separate text boxes, username at the top and password below it, they will then select a button below the password text box which will allow them to proceed to the home screen if their password and username are valid. Otherwise, they will be able to press a button, in the bottom right-hand section of the screen, to register as a new user, this will direct them to a sign-up screen or they may select a help button in the top right-hand corner of the screen.

Screen 2 (sign-up): This screen will contain the school logo in the top left-hand corner and will allow a new student account to be created by entering a username into a single text box and a password twice, each time into a separate text box, they will also be required to enter their full name, grade, class, gender (2 options as radio buttons) and select which portfolios they are a part of, using text boxes for all fields excepting check boxes for the portfolios and radio buttons for gender. They will be able to press a button at the bottom centre of the screen to sign-up provided all of the information has been entered. Otherwise, they will be able to access the help button in

the top right-hand corner of the screen.

Screen 3 (Home): The home page will allow students to view their details and select a button to edit their details or they may navigate through tabs found at the top of the screen to other areas of the program. They will be presented with the school logo in the top left-hand corner, the help button in the top right-hand corner, beneath this, in the centre of the screen will be their name, in , their student ID to the right of their name, beneath this will be their grade, class and gender. Their portfolios will be at the lowest section of the text box, with their total hours for each portfolio next to the portfolio name.

Edit Account (Student): This will allow the learner to edit their account details, they will be presented with a title at the top of the page, “Edit your account”, beneath the title will be labels naming the field to the left of text fields, each beneath the previous with their first name, surname, grade and class, below this will be radio buttons for gender and then check boxes for the portfolio selection. At the very bottom of the screen will be an “enter” button and to the left will be a “cancel” button. Should the user press the “Enter” button with invalid information selected, error labels will appear to the right of any field with invalid information.

Screen 4 (Events): This will allow learners to view events related to the service portfolios they are involved in. They will be presented with the help button in the top right-hand corner, the table will be to the left of the screen, containing the event ID, event name, event date, portfolio and the number of hours the event counts for. There will be a date picker, 2 text boxes and 3 buttons to the right of the table, the text boxes will each provide a separate search function (Date, name, and ID), they will be ordered as a text box above a button, in three sets.

Screen 5 (Participation): Will allow Students to see which events they have logged their participation in. They will be presented with the help button in the top right-hand corner, the table will be to the left of the screen, containing their student ID, event ID, event name, portfolio and number of hours. There will be 3 text boxes and 3 buttons to the right of the table, the text boxes will each provide a separate search function (Date, name, and ID), they will be ordered as a text box above a button, in three sets. Beneath the table will be a text box and a create button, to the left of which there will be a text box and a delete button, where students can enter an event ID to delete or create a record of their participation in an event.

Screen 6 (Home admin): The home page will allow admins to see their information, select a button to edit it, select a button to create a new admin account or they can navigate to other areas of the program through use of tabs, found in the top left of the screen. They will be presented with the school logo in the top left-hand corner, below the tabs, the help button in the top right-hand corner, beneath this, in the centre of the screen will be their name, to the left of which will be their ID, beneath their name will be their portfolios. The edit account button will be below the text box containing their information, to the right of this button will be the create new admin account button.

Edit Account (Admin): This will allow the Admin to edit their account details, they will be presented with a title at the top of the page, “Edit your Admin account”, beneath the title will be labels naming the field to the left of text fields, each beneath the previous with their first name, surname and then check boxes for the portfolio selection. At the very bottom of the screen will be an “enter” button and to the left will be a “cancel” button. Should the user press the “Enter” button with invalid information selected error labels will appear to the right of any field with invalid information.

Create new Account (Admin): This will allow the Admin to create new admin accounts, they will be presented with a title at the top of the page, “Create new Admin account”, beneath the title will be labels naming the field to the left of text fields, each beneath the previous with their full name, username, password, confirm password and then check boxes for the portfolio selection. At the

very bottom of the screen will be an “enter” button and to the left will be a “cancel” button. Should the user press the “Enter” button with invalid information selected error labels will appear to the right of any field with invalid information.

Screen 7 (Participation admin): Will allow admins to see events in which students have logged their participation. They will be presented with the help button in the top right-hand corner, the table will be to the left of the screen, containing the event ID, event name, student ID, Student name, portfolio, and number of hours. There will be 4 text boxes and 4 buttons to the right of the table, the text boxes will each provide a separate search function (Portfolio, student name, event name, event ID), they will be ordered as a text box above a button, in four sets. Beneath the table will be two text boxes and a create button, to the left of which will be another two text boxes and a delete button where admin can enter an event ID and student ID to delete or create a record of participation in an event.

Scree 8 (Events admin): Allows admins to see all events related to their portfolio(s) and select a button to add new events. They will be presented with the help button in the top right-hand corner, the table will be to the left of the screen, containing the event ID, event name, event date, portfolio, and the number of hours for which the event counts. There will be a date picker, 2 text boxes and 3 buttons to the right of the table, the text boxes will each provide a separate search function (Date, name, and ID), they will be ordered as a text box above a button, in three sets. At the bottom of the screen, to the left there will be a button which will direct the admin to a screen where they can create a new event. To the right there will be another button which will direct the admin to a page where they can edit an event’s details.

Create new event (Admin): This will allow the Admin to create new events, they will be presented with a title at the top of the page, “New Event”, beneath the title will be labels naming the field to the left of text fields, each beneath the previous with the event name, hours, Date (date picker) and then radio buttons where a portfolio choice must be made (only the portfolios the admin participates in will be shown). At the very bottom of the screen will be an “enter” button and to the left will be a “cancel” button. Should the user press the “Enter” button with invalid information selected error labels will appear to the right of any field with invalid information.

Edit event (Admin): This will allow the Admin to edit events, they will be presented with a title at the top of the page, “Edit Event”, beneath the title will initially be a single label and text field where the user must enter an event ID, to the right of this will be a “find event” button and at the bottom of the screen will be a cancel button, once a valid ID has been entered, the find event button will disappear and the rest of the labels and text fields will appear, labels naming the field to the left of text fields, each beneath the previous with the event name, hours, Date (date picker) and then radio buttons where a portfolio choice must be made (only the portfolios the admin participates in will be shown). At the very bottom of the screen will be an “enter” button and to the right will be a “cancel” button. Should the user press the “Enter” button with invalid information selected error labels will appear to the right of any field with invalid information.

Screen 9 (Students): Allow admins to see all students that participate in their portfolio(s). They will be presented with the help button in the top right-hand corner, the table will be to the left of the screen, containing the student ID, Student name, grade, class, gender, and their total hours for each portfolio. There will be 3 text boxes and 3 buttons to the right of the table, the text boxes will each provide a separate search function (name, grade, and ID), they will be ordered as a text box above a button, in 3 sets.

Screen 10 (Help): available from every main screen of the program to help users overcome any problems they may be having. The user will be presented with a cross in the top right-hand corner, which will let them return to the page they came from. The main section of the page will contain

many buttons, each providing a video solution to a problem the user may be having. At the bottom of the page there will be an email address which will allow the user to access a help desk in the case that none of the provided solutions are helpful.		
0 – User Interface not specified or incorrectly specified	1 – One or two items are inadequately specified	2 – User Interface completely specified

Specification of help

There will be a separate page dedicated to help. This page will include questions (on buttons) about problems the user may be experiencing and will provide them with links to videos (Available on YouTube) that will explain, in detail, how to perform each task or overcome the problem. Should none of these solutions be helpful to the user, an Email address with be listed at the bottom of the page so that they can contact someone who will be able to assist them in overcoming the issue. The user will be able to navigate to this page from anywhere in the program, by clicking the “?” button in the top right-hand corner of any page.

The available solutions will be provided for the following questions:

- How do I log in?
- How do I sign up?
- How do I create an admin account?
- How do I change my portfolios?
- How do I search for an event?
- How do I create an event? (Admin only)
- How do I edit an event? (Admin only)
- How do I search for a participation record?
- How do I change a participation record?
- How do I delete a participation record?
- How do I search for a student? (Admin only)
- Etc.

0 – Help not discussed.	1 - Help partially discussed with omissions and/or errors. Help is not appropriate for the type of project (e.g. printed manual is not required for mobile applications).	2 - Help completely described and a variety of help is described. Storage of help related data has been specified. Help provision appropriate for application type.

Specifications of Data Storage

For each entity which your program will manage, list the data which will need to be stored (**not** how it will be stored). Give a very brief description for each item listed.

Database Tables:

Students:
StudentID [number] – uniquely identify each student by number
FirstName [String] – The student's first name
Surname [String] – The student's surname
Grade [number] – The grade that the student is currently in
Class [number] – The class that the student is currently in
Gender [char] – M/F to indicate the gender of the student
Catering [Boolean] – whether the student does catering
Photography [Boolean] – whether the student does photography
AV[Boolean] – whether the student does AV
First Aid [Boolean] – whether the student does First Aid

Admin:
AdimnID [number] – uniquely identify each admin by number
FirstName [String] – The admin's first name
Surname [String] – The admin's surname
Catering [Boolean] – whether the admin assists with catering
Photography [Boolean] – whether the admin assists with photography
AV[Boolean] – whether the admin assists with AV
First Aid [Boolean] – whether the admin assists with First Aid

Events:
EventID [number] – uniquely identify each event by number
EventName [String] – The name/title of the event
EventDate [Date] – The date of the event
Portfolio [String] – Which portfolio the event relates to
Hours [Number] – the number of hours the event counts for

Participation:
EventID [number] – uniquely identify each event by number
StudentID [number] – uniquely identify each student by number

Records are created:
- When a student signs up, they enter their information, and that information is used to create a record in the student table
- When an admin creates another admin account the information is used to create a record in the admin table
- When an admin creates an event, the information entered is used to create a record in the events table
- When a student logs their participation, the information entered is used to create a record in the participation table
- When an admin creates a participation record, the information entered is used to create a record in the participation table.

Records are accessed:
- On the home screen when the user's information is shown in a text field
- In the events section of the application, learners and teachers can view events related to their portfolio selections.
- In the participation section of the application, learners can view their own participation records.
- In the participation section of the application, admins can view all participation records

related to their selected portfolios.
- Admins can view the student records in the students section of the application.

Records are updated:

- Students can update their account details on the edit account page.
- Admins can update their account details on the edit account page.
- Admins can edit existing events in the events section of the application.

Records are deleted:

- Admins can delete participation records in the participation section of the application.
- Students can delete participation records in the participation section of the application.

Text File (Notepad)

Each line will contain a username and password in the following format:
Username#Password#(Admin/Student)

0 – No information given on the data to be stored	1 – Only a few items are incorrectly described	2 – Many items are described with a few errors	3 – All data to be stored has been correctly described

Hardware and Software Requirements

List what hardware and software you will need to **develop** your program. You must give the full name and version of each program you will require and a brief description for what you will use the software.

Programmer:

Software:

- Windows 10 operating system
- Java 8 or higher
- NetBeans 12
- MS Access 365

Hardware:

- Processor: 1 gigahertz (GHz) or faster processor
- RAM: Minimum 4GB
- Hard disk space: Minimum 30GB

User requirements:

Software:

- Windows 10 operating system
- Java 8 or higher
- MS Access 365

Hardware:

- Processor: 1 gigahertz (GHz) or faster processor
- RAM: Minimum 3GB
- Hard disk space: Minimum 25GB

0 – Hardware and Software not discussed	1 – Hardware and software is partially discussed for development, includes detail for what software is needed for what task	2 – Hardware and software is completely discussed for development, software list includes versions